

FIT5225 2026 Assignment 1: Cloud Eco			
Marking Feedback Sheet			
Student Name : Bharath Gandhimani (35501308)			
Marked By : Marker Name			
Sections	Criteria	Comments	Mark Awarded
Section 1: Web Service API Engineering (20 Marks)	D: The APIs are highly robust. Image encoding and decoding are handled efficiently using optimised libraries. The application architecture demonstrates a clean separation of concerns, isolating routing logic from machine-learning inference logic. Error handling for invalid JSON payloads is present.	Comprehensive inline comments and code documentation were missing; these need to be added to ensure code maintainability and clarity.	14
Section 2: Containerization and Optimisation (10 Marks)	HD: Flawless, production-grade Dockerfile. The student demonstrates advanced techniques such as multi-stage builds to compile dependencies separately from the final runtime image. The final image footprint is drastically minimised, and the container is configured to execute as a non-root user for enhanced security compliance.		8
Section 3: Infrastructure as Code (IaC) (10 Marks)	C: IaC scripts are provided, but they only provision a fraction of the required infrastructure. The script's execution does not require manual intervention.	Docker build and push was not automated.	6
Section 4: Kubernetes Deployment (15 Marks)	D: Flawless deployment architecture. The Service correctly distributes traffic. Replica sets are easily scalable via configuration changes. The YAML files adhere strictly to Kubernetes best practices, utilising explicit label selectors and appropriate namespace management.	The utilization of a private image registry was missing; the Docker image was pulled from a public repository, and a transition to a private repository (configured with ImagePullSecrets) needs to be done to ensure container security.	10.5
Section 5: Load Generation and Performance Baseline (20 Marks)	D: The system meets the required performance baseline (sustaining 5 to 10 QPS per 1 vCPU pod) with low error rates and acceptable latency, demonstrating adequate application-level engineering and event-loop management.	The implementation needs to transition away from hardcoded test images to allow for dynamic benchmarking during the load test.	14
Section 6: Empirical Benchmark Report (10 Marks)	D: In-depth empirical analysis. The student accurately identifies specific computational bottlenecks. Graphs are professionally rendered with correct axes, units, and descriptive legends.	A transition toward publication-quality analysis and formatting needs to be done to achieve the highest marking tier.	7
Section 7: Interview Performance (15 Marks)	D: The student navigates their system architecture with high confidence. They provide clear, logical justifications for their architectural decisions and demonstrate a strong, practical understanding of Docker optimisation and Kubernetes networking routing.		10.5
	Subtotal:		70
	Late Submission Day(s):	0	0
	Final Grade:	D	70
	Contribution to Final Marks:		21